

DeepSeek-V2

Efficient MoE Language Model with Multi-Head Latent Attention

236B total params · 21B activated · 128K context

236B

Total Parameters

21B

Activated / Token

128K

Context Length

42.5%

Training Cost ↓

The LLM Scaling Dilemma

Performance vs. Training Cost and Inference Speed

Prohibitive Training Cost

Dense LLMs require full activation of all parameters per forward pass. Training 70B+ dense models demands enormous GPU hours and memory.

KV Cache Bottleneck

Standard Multi-Head Attention stores full K/V per head per token. For 128K context, cache size grows linearly, crippling inference latency.

Slow Generation Throughput

Dense activation means every token triggers computation across the entire model — limiting throughput and increasing per-token cost.

DeepSeek-V2 Architecture

MLA + DeepSeekMoE in a Transformer Block

Multi-Head Latent Attention (MLA)

- Low-rank KV joint compression
- Decoupled RoPE for positional encoding
- Cached: compressed latent vector c_t^{KV}
- On-the-fly reconstruction of K, V, RoPE-K
- Better than MHA with far less cache

DeepSeekMoE FFN

- Fine-grained expert segmentation
- Shared experts (N_s): always activated
- Routed experts (N_r): top- K_r sparse activation
- Device-limited routing for comm. efficiency
- Auxiliary load-balancing losses

Multi-Head Latent Attention (MLA)

Low-Rank KV Joint Compression Reduces KV Cache by 93.3%

Core Innovation: Low-Rank Compression

Standard MHA

Caches full K and V per head per token — cache grows with context length and head count.

MLA Solution

Compresses K and V into a compact latent vector c_t^{KV} during inference.

Decoupled RoPE

Adds a separate positional key (RoPE-K) decoupled from the main attention computation.

Reconstruction

Full K, V, and RoPE-K are reconstructed on-the-fly via learned projection matrices.

Attention Mechanisms vs. KV Cache Size

Method	Cache/Head	Performance
MHA	Full K+V per token	Baseline
GQA	Reduced K+V (grouped)	Lower
MQA	Single K+V shared	Lowest
MLA ★	Compressed latent c_t^{KV}	Highest

★ MLA achieves BETTER performance than MHA while reducing KV cache by 93.3%.

93.3% KV Cache Reduction

vs. standard MHA — enables 128K context at manageable memory

DeepSeekMoE

Fine-Grained Experts with Shared Expert Isolation

Shared Experts (N_s)

Always activated.
Capture common knowledge and general language patterns.
Never dropped.

Routed Experts (N_r)

Sparsely activated via top- K_r router.
Specialized knowledge experts.
Each token activates K_r out of N_r .

Fine-Grained Segmentation

Smaller, more specialized experts increase specialization potential vs. coarse-grained MoE.

Top-K Routing

Each token routed to K_r out of N_r routed experts based on affinity scores. Sparse activation = efficient.

Device-Limited Routing

Each token limited to experts on $\leq M$ devices, reducing cross-node communication during training.

Auxiliary Load-Balancing

Expert+device auxiliary losses prevent token collapse and ensure balanced compute.

Token Dropping (Training)

Training-only strategy prevents expert overflow; dropped tokens do not affect inference.

Efficiency Gains vs. DeepSeek 67B

42.5% Training Cost Reduction · 5.76× Generation Throughput

42.5%

Training Cost
Reduction

vs. DeepSeek 67B

93.3%

KV Cache
Reduction

vs. DeepSeek 67B

5.76×

Generation
Throughput

vs. DeepSeek 67B

MMLU Performance: Pareto-Optimal Efficiency

Top MMLU Among Open-Source Models at Only 21B Activated Parameters



DeepSeek-V2 Chat (RL)

SFT (1.5M sessions) + GRPO RL → Strong Multilingual Alignment

Pretraining

8.1T tokens
Multi-source corpus
Extended Chinese

SFT

1.5M conversational
sessions · 监督微调

GRPO RL

Group Relative
Policy Optimization

Chat (RL)

Final aligned
model

Chat Evaluation Scores

AlpacaEval 2.0
(length-controlled win rate)

38.9%

Outperforms most open-source
and many closed-source models

MT-Bench
(overall)

8.97

Competitive with
top closed-source models

AlignBench
(Chinese, overall)

7.91

Best open-source model in Chinese;
rivals closed-source systems

Key Takeaways

Architectural Innovation Unlocks Efficient LLM Scaling

Multi-Head Latent Attention (MLA)

Superior to MHA — reduces KV cache by 93.3% with better performance.
Strictly better than GQA/MQA alternatives.

DeepSeekMoE

Fine-grained experts + shared expert isolation enable training
236B params at 42.5% lower cost than a 67B dense model.

Pareto-Optimal Performance

21B activated params matches/exceeds 70B-class dense models
(LLaMA 3 70B, Qwen1.5 72B) on MMLU (78.5).

Strong Multilingual Alignment

DeepSeek-V2 Chat (RL) is the strongest open-source chat model
in Chinese (AlignBench 7.91) and competitive in English.

Open-Source Availability

Full weights and training details released — enabling
efficient deployment at a fraction of the compute cost.